

The styrene production process begins with an equi-molar feed of benzene and ethylene, where the

molar composition of the inlet stream is 0.5 for Benzene and 0.5 for Ethylene. This feed, initially preheated to

95°C at a constant pressure of 1.01325 bar, has a molar flow rate of 9.419034 kmol/h and a mass flow rate of 500 kg/h. The stream is then sent to a

pump to increase its pressure from 1.01325 bar to 13 bar. Following this, the pressurized feed enters a

conversion reactor, where ethyl benzene is produced with a 70% conversion rate based on benzene. The product stream from this reactor, containing ethyl benzene, unreacted ethylene, and benzene, is then directed to a

cooler that reduces its temperature to 150°C. After cooling, the stream proceeds to a

first distillation column, which separates purified ethyl benzene as the bottom product from the distillate containing unreacted ethylene and benzene.